Open-RMF demos

Demonstrations of the Open-RMF

Pre-requisites

Docker

https://docs.docker.com/engine/install/ubuntu/

OSRF rocker

https://github.com/osrf/rocker

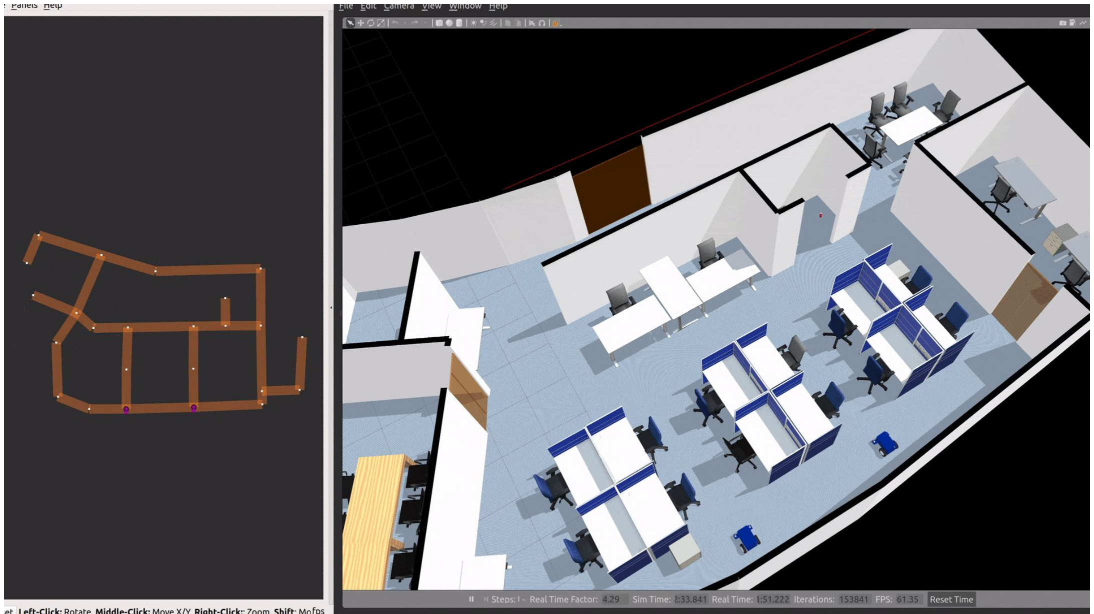
Pull RMF image

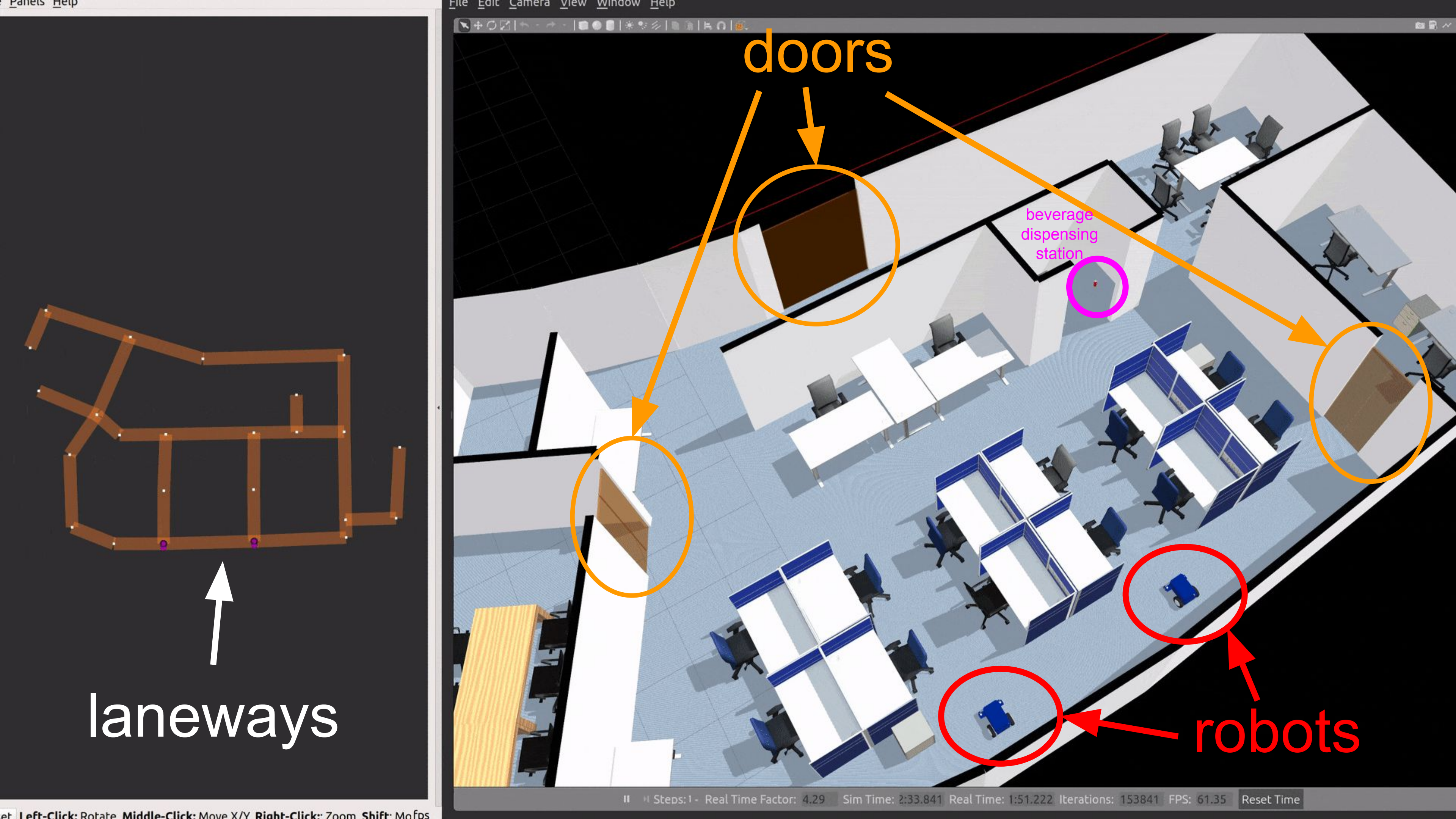
https://github.com/open-rmf/rmf/pkgs/container/rmf%2Frmf_demos

```
docker pull ghcr.io/open-rmf/rmf/rmf_demos:jazzy-rmf-latest
```

Office World

An indoor office environment for robots to navigate around. It includes a beverage dispensing station, controllable doors and laneways which are integrated into RMF. We will showcase 2 types of Tasks: Delivery and Patrol





Panels Help
File Edit Camera View Window Help
doors
beverage dispensing station
laneways
robots
Left-Click: Rotate Middle-Click: Move X/Y Right-Click: Zoom Shift: Mo
Steps: 1 Real Time Factor: 4.29 Sim Time: 2:33.841 Real Time: 1:51.222 Iterations: 153841 FPS: 61.35 Reset Time

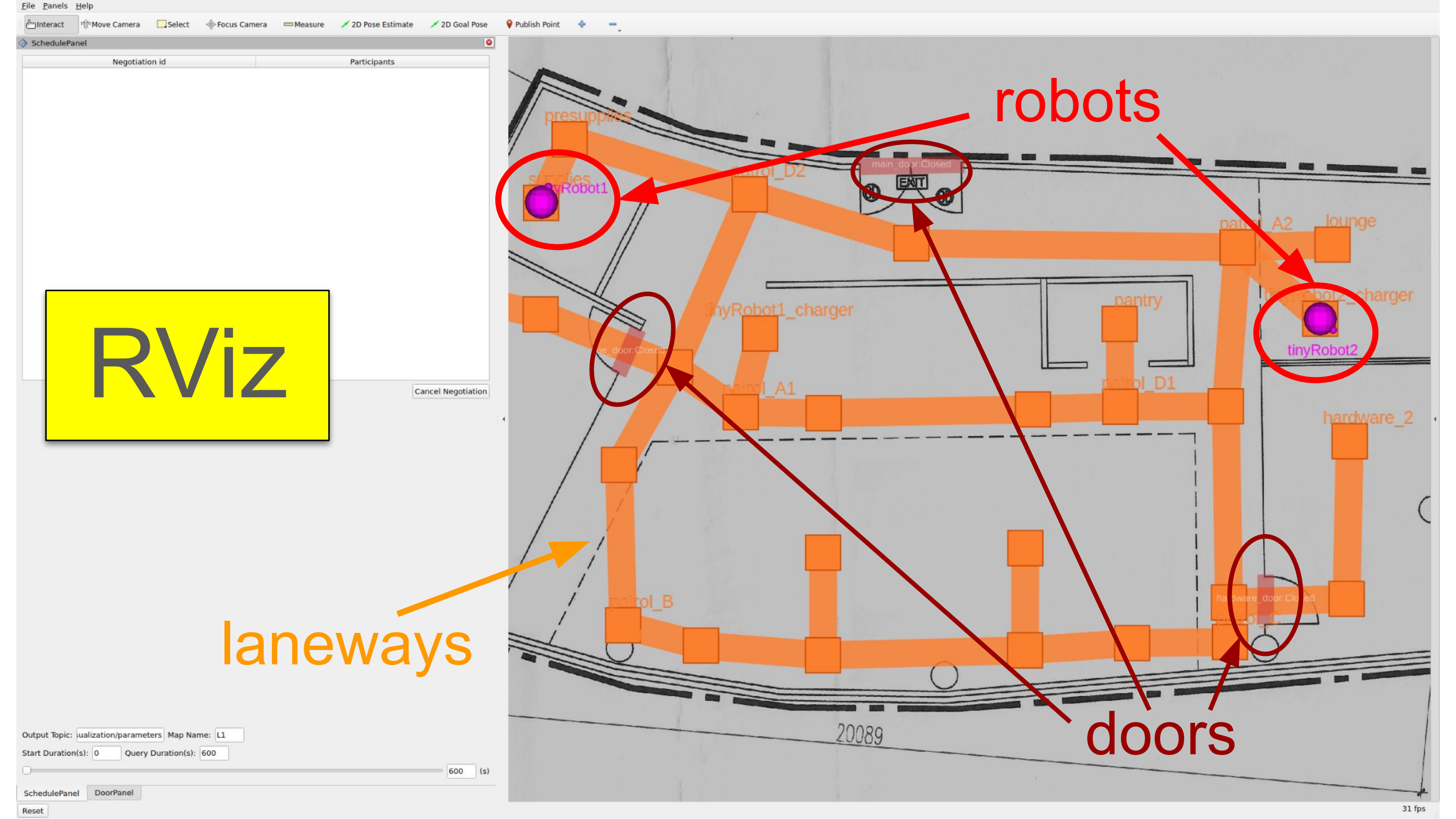
Terminal 1

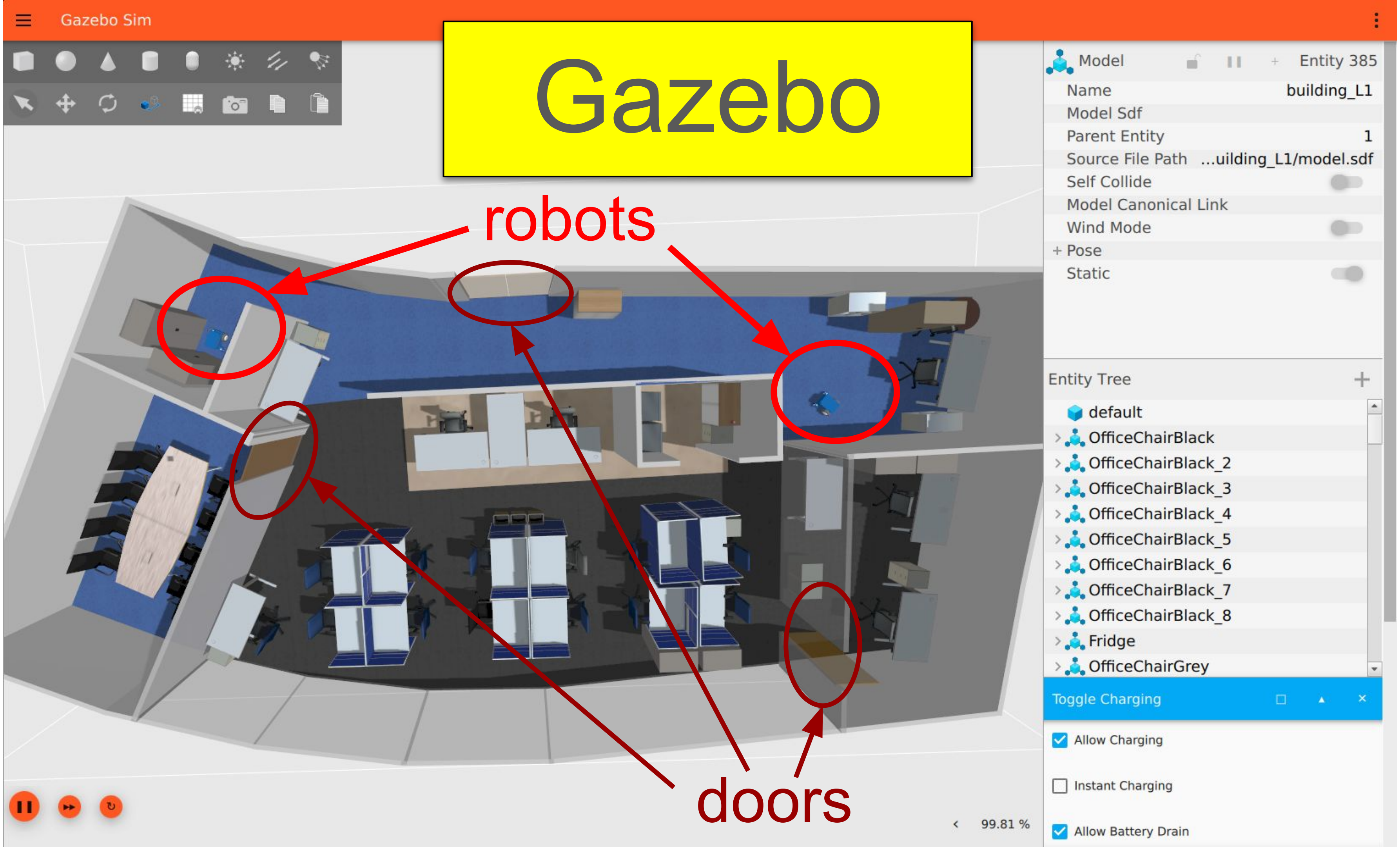
GPU & GUI options

```
rocker --nvidia --x11 \
  ghcr.io/open-rmf/rmf/rmf_demos:jazzy-rmf-latest \
    ros2 launch rmf_demos_gz \
      office.launch.xml
```

ROS command

Container image


File Panels Help
Interact
Move Camera
Select
Focus Camera
Measure
2D Pose Estimate
2D Goal Pose
Publish Point
SchedulePanel
Negotiation id
Participants
Cancel Negotiation
RViz
laneways
robots
doors
presupplies
tinyRobot1
main_door:Closed
EXIT
lounge
tinyRobot1_charger
pantry
patrol_A1
patrol_D1
tinyRobot2
hardware_2
patrol_B
20089
Output Topic:
ualization/parameters
Map Name:
L1
Start Duration(s):
0
Query Duration(s):
600
600
(s)
SchedulePanel
DoorPanel
Reset
31 fps


Gazebo Sim
Gazebo
robots
doors
Model
Entity 385
Name
building_L1
Model Sdf
Parent Entity
1
Source File Path
...uilding_L1/model.sdf
Self Collide
Model Canonical Link
Wind Mode
Pose
Static
Entity Tree
default
OfficeChairBlack
OfficeChairBlack_2
OfficeChairBlack_3
OfficeChairBlack_4
OfficeChairBlack_5
OfficeChairBlack_6
OfficeChairBlack_7
OfficeChairBlack_8
Fridge
OfficeChairGrey
Toggle Charging
Allow Charging
Instant Charging
Allow Battery Drain
99.81 %

Terminal 2

You can request the robot to deliver a can of coke from **pantry** to **hardware_2** through the following:

```
docker run --rm -it \
  ghcr.io/open-rmf/rmf/rmf_demos:jazzy-rmf-latest \
    ros2 run rmf_demos_tasks dispatch_delivery \
      -p pantry     -ph coke_dispenser \
      -d hardware_2 -dh coke_ingestor \
      --use_sim_time
```

Container image

ROS command

dispatch_task_request

Json msg payload:


```
{
  "type": "dispatch_task_request",
  "request": {
      "unix_millis_request_time": 0,
      "unix_millis_earliest_start_time": 0,
      "requester": "rmf_demos_tasks",
      "category": "delivery",
      "description": {
        "pickup": {
          "place": "pantry",
          "handler": "coke_dispenser",
          "payload": []
        },
        "dropoff": {
          "place": "hardware_2",
          "handler": "coke_ingestor",
          "payload": []
        }
      }
  }
}
```


Got response:


```
{
  'state': {
    'booking': {
      'id': 'delivery.dispatch-a2d7210bec',
      'requester': 'rmf_demos_tasks',
      'unix_millis_earliest_start_time': 0,
      'unix_millis_request_time': 0
    },
    'category': 'delivery',
    'detail': {
      'dropoff': {
        'handler': 'coke_ingestor',
        'payload': [],
        'place': 'hardware_2'
      },
      'pickup': {
        'handler': 'coke_dispenser',
        'payload': [],
        'place': 'pantry'
      }
    },
    'dispatch': {'errors': [], 'status': 'queued'},
    'status': 'queued',
    'unix_millis_start_time': 0
  },
  'success': True
}
```

Terminal 2

You can also request the robot to move back and forth between **coe** and **lounge** through the following:

```
docker run --rm -it \
 ghcr.io/open-rmf/rmf/rmf_demos:jazzy-rmf-latest \
  ros2 run rmf_demos_tasks dispatch_patrol \
    -p coe lounge -n 3 --use_sim_time
```

Container image

ROS command

dispatch_task_request

Json msg payload:

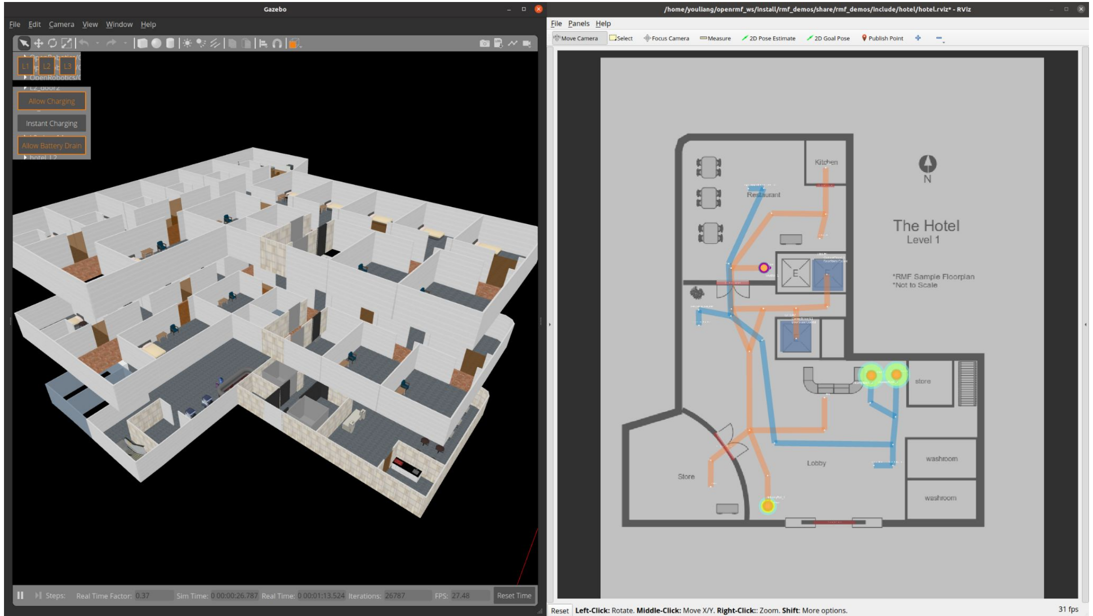

```
{
  "type": "dispatch_task_request",
  "request": {
    "unix_millis_request_time": 0,
    "unix_millis_earliest_start_time": 0,
    "requester": "rmf_demos_tasks",
    "category": "patrol",
    "description": {
      "places": [
        "coe",
        "lounge"
      ],
      "rounds": 3
    }
  }
}
```

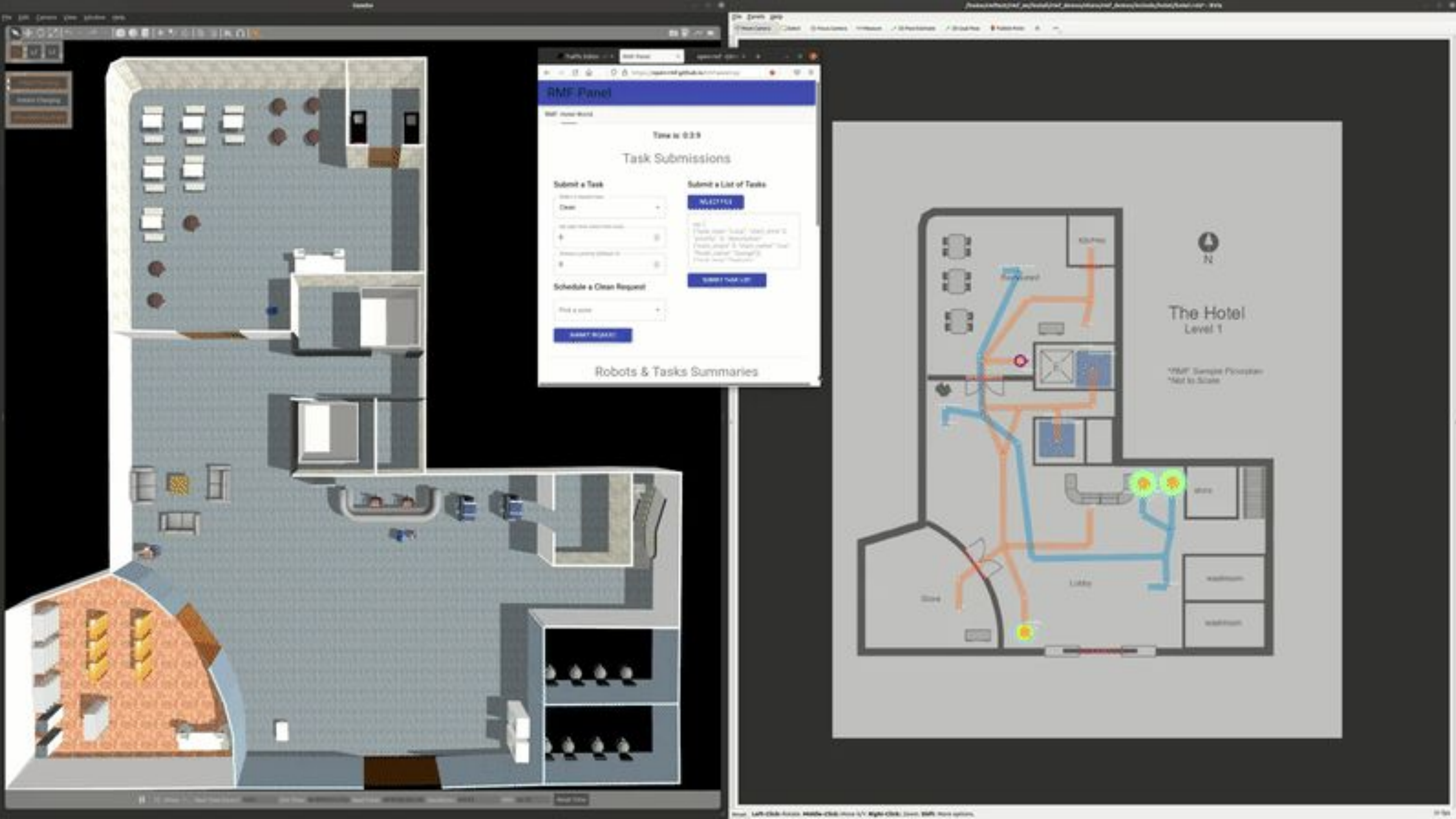

Got response:


```
{
  'state': {
    'booking': {
      'id': 'patrol.dispatch-2d97be1dcc',
      'requester': 'rmf_demos_tasks',
      'unix_millis_earliest_start_time': 0,
      'unix_millis_request_time': 0
    },
    'category': 'patrol',
    'detail': {
      'places': [
        'coe',
        'lounge'
      ],
      'rounds': 3
    },
    'dispatch': {'errors': [], 'status': 'queued'},
    'status': 'queued',
    'unix_millis_start_time': 0
  },
  'success': True
}
```

Hotel World

This hotel world consists of a lobby and 2 guest levels. The hotel has two lifts, multiple doors and 3 robot fleets (4 robots). This demonstrates an integration of multiple fleets of robots with varying capabilities working together in a multi-level building.

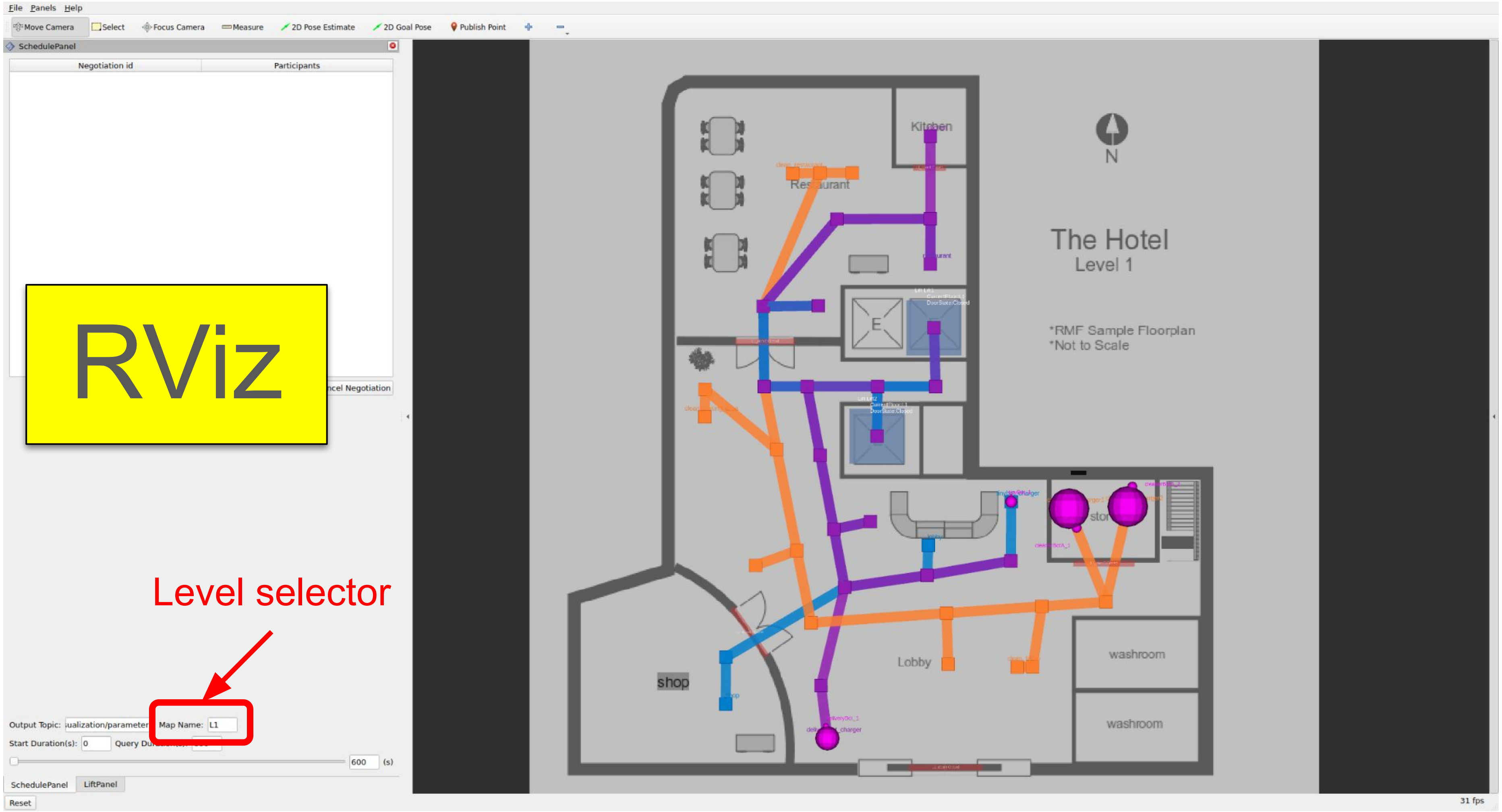


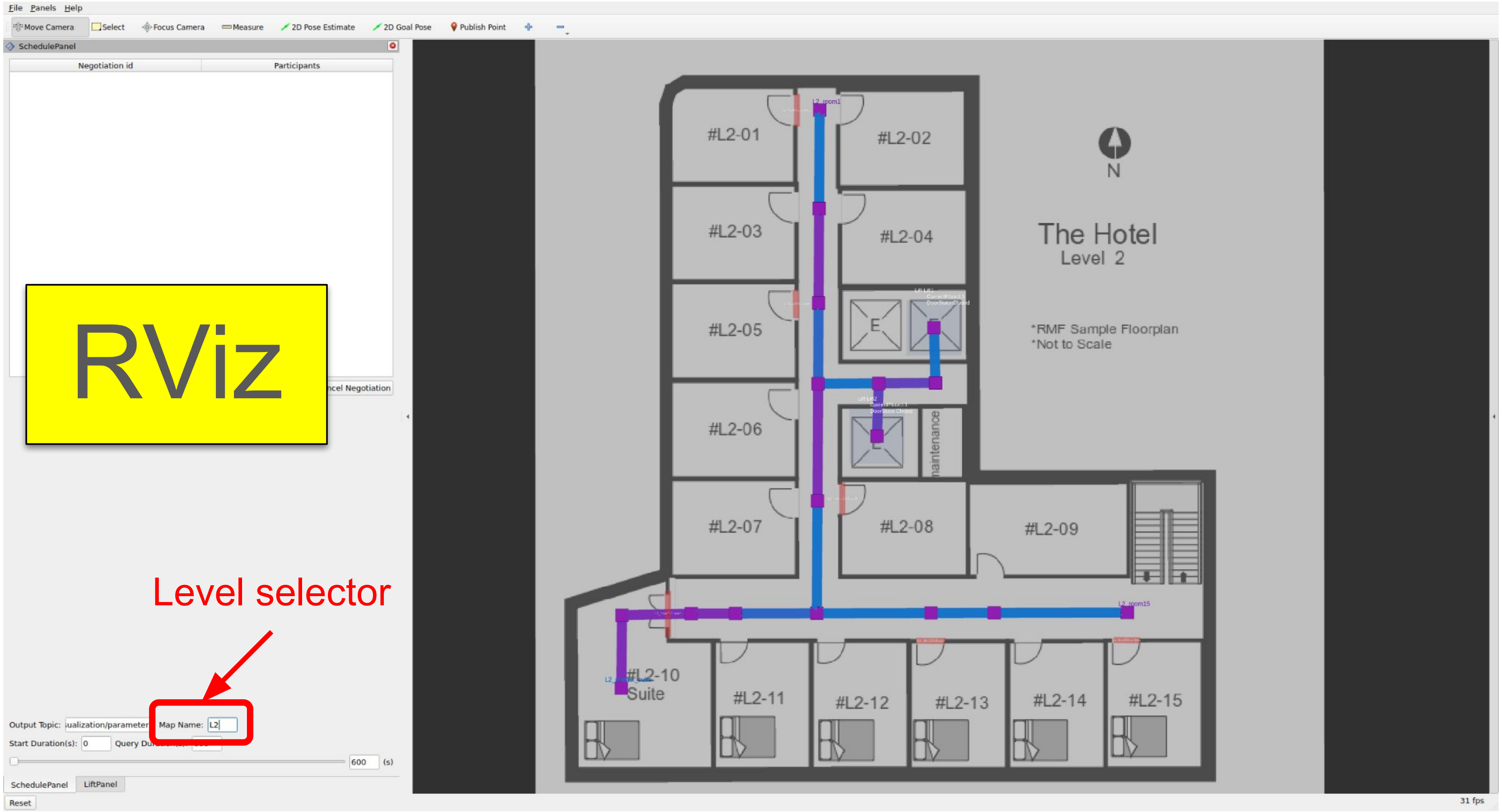


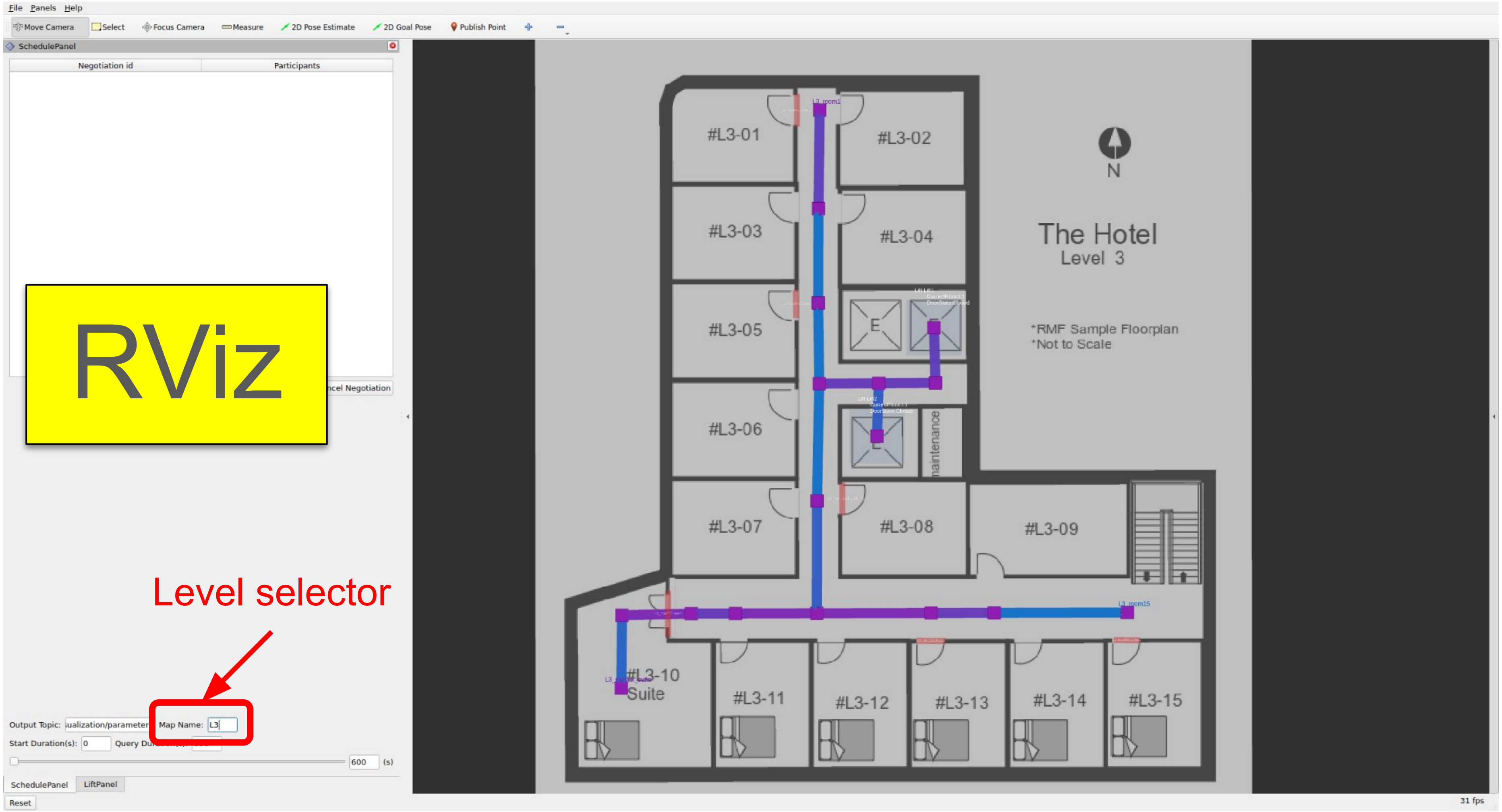
RMF Panel
Task Submissions
Submit a Task
Submit a List of Tasks
Schedule a Clean Request
Robots & Tasks Summaries
The Hotel
Level 1
N
Lobby
Store

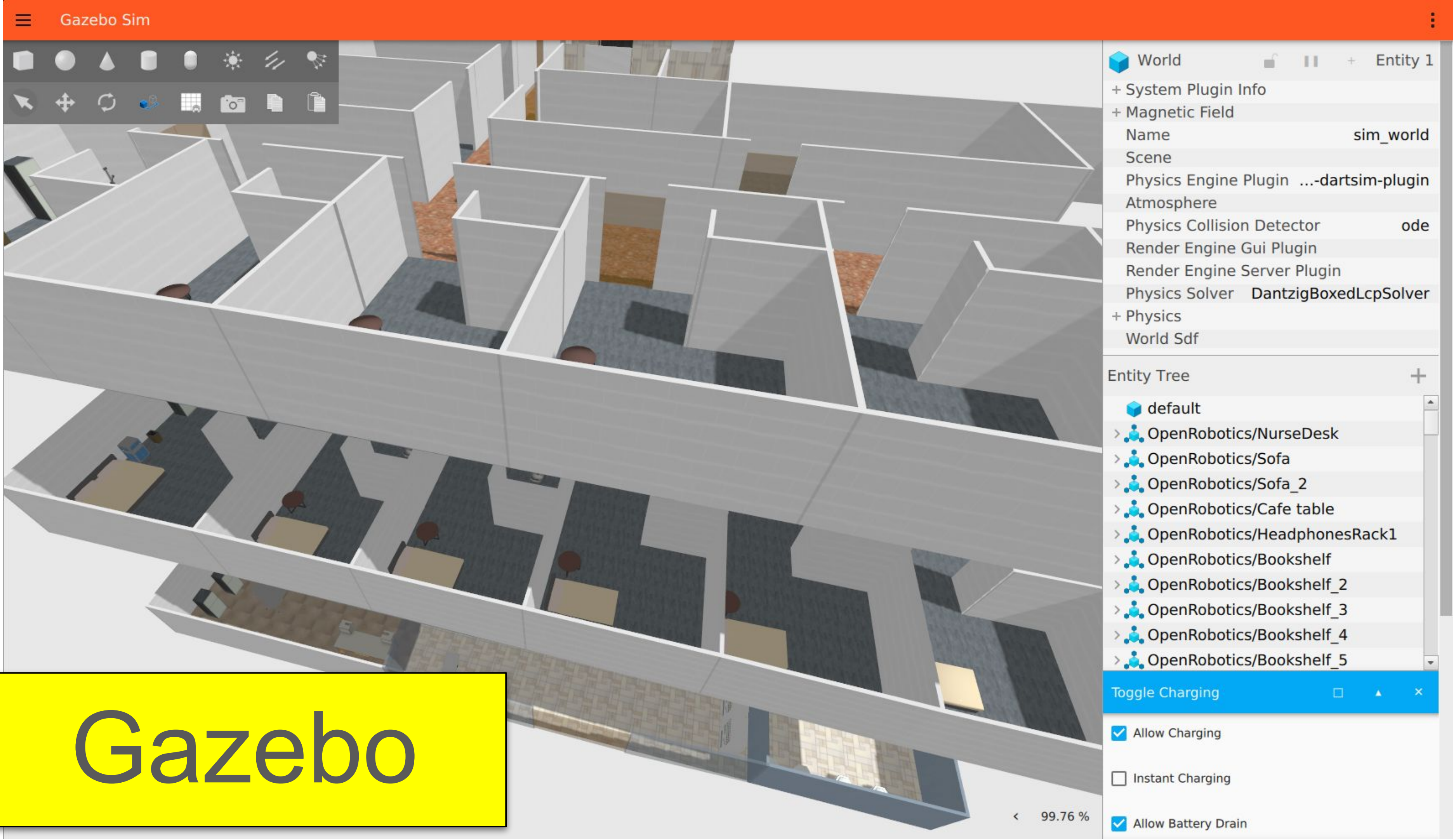
Terminal 1

```
rocker --nvidia --x11 \
  ghcr.io/open-rmf/rmf/rmf_demos:jazzy-rmf-latest \
    ros2 launch rmf_demos_gz \
      hotel.launch.xml
```

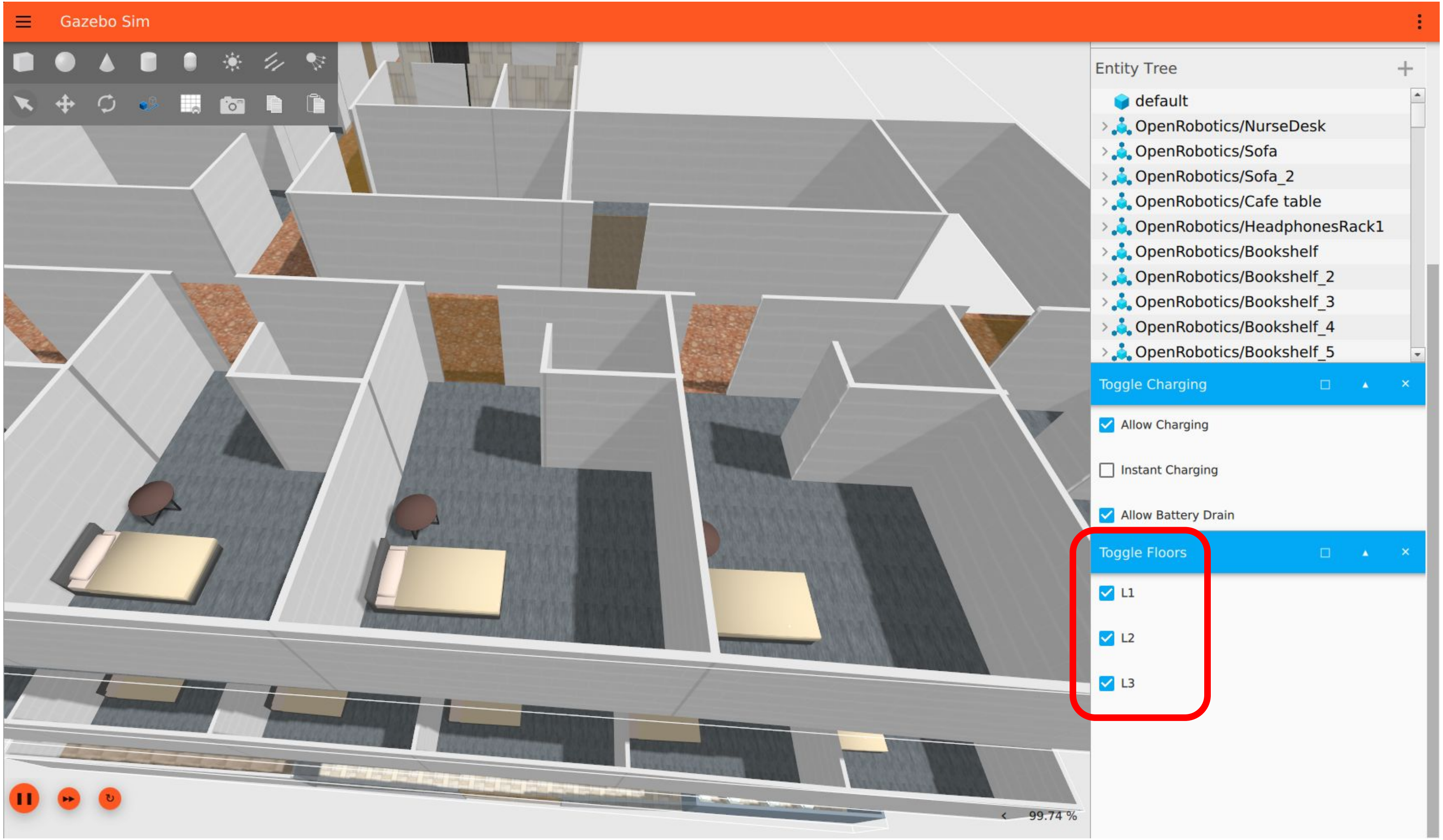

File Panels Help
Move Camera
Select
Focus Camera
Measure
2D Pose Estimate
2D Goal Pose
Publish Point
SchedulePanel
Negotiation id
Participants
RViz
ncel Negotiation
Level selector
Output Topic: ualization/parameter
Map Name: L1
Start Duration(s): 0
Query Du
600 (s)
SchedulePanel
LiftPanel
Reset
31 fps
Kitchen
Restaurant
N
The Hotel
Level 1
*RMF Sample Floorplan
*Not to Scale
E
Lobby
shop
washroom
washroom

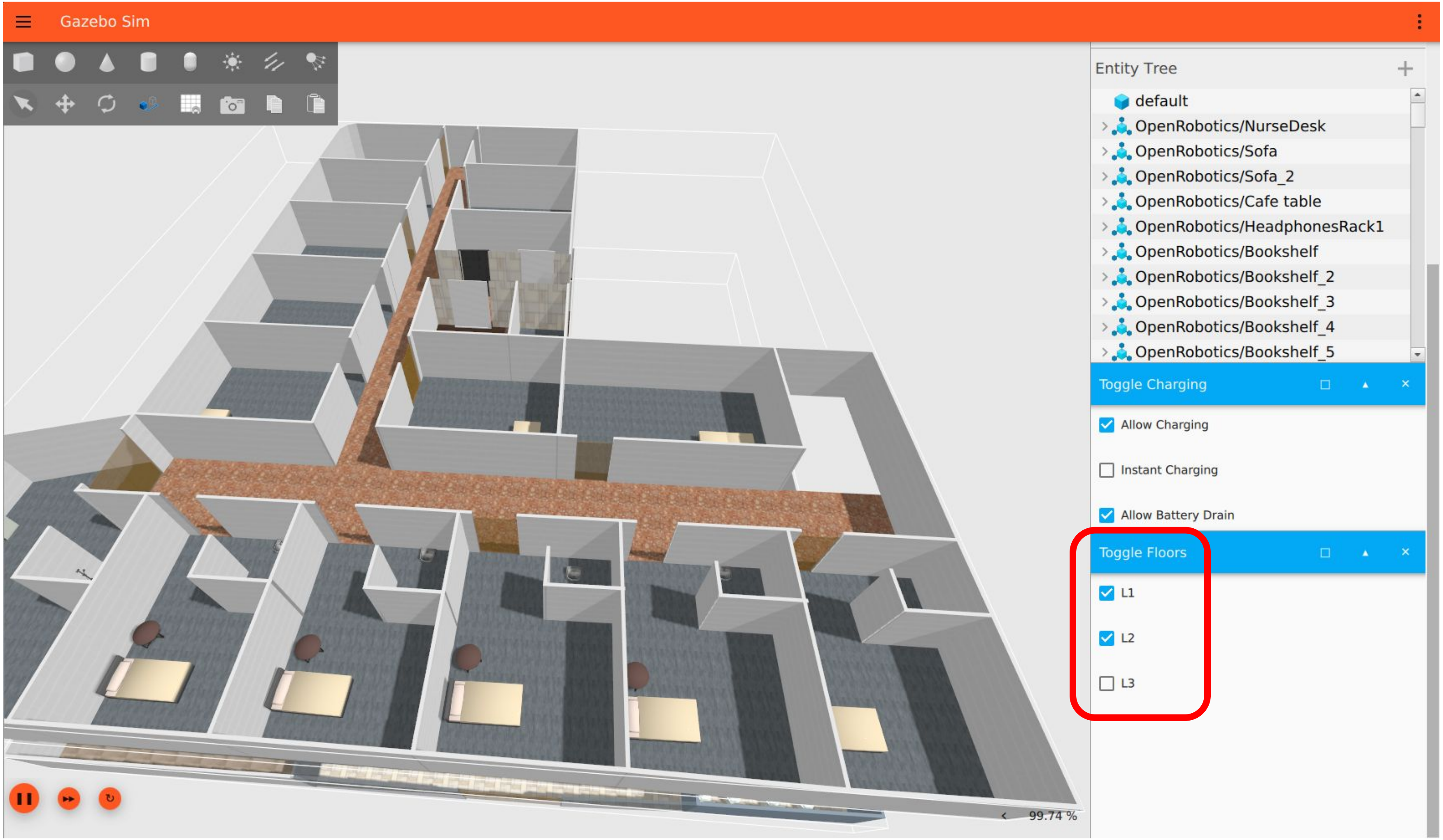

RViz
Level selector
File Panels Help
Move Camera
Select
Focus Camera
Measure
2D Pose Estimate
2D Goal Pose
Publish Point
SchedulePanel
Negotiation id
Participants
ncel Negotiation
Output Topic: ualization/parameter
Map Name: L2
Start Duration(s): 0
Query Du
600
(s)
SchedulePanel
LiftPanel
Reset
31 fps
#L2-01
#L2-02
#L2-03
#L2-04
#L2-05
#L2-06
#L2-07
#L2-08
#L2-09
#L2-10
Suite
#L2-11
#L2-12
#L2-13
#L2-14
#L2-15
maintenance
N
The Hotel
Level 2
*RMF Sample Floorplan
*Not to Scale

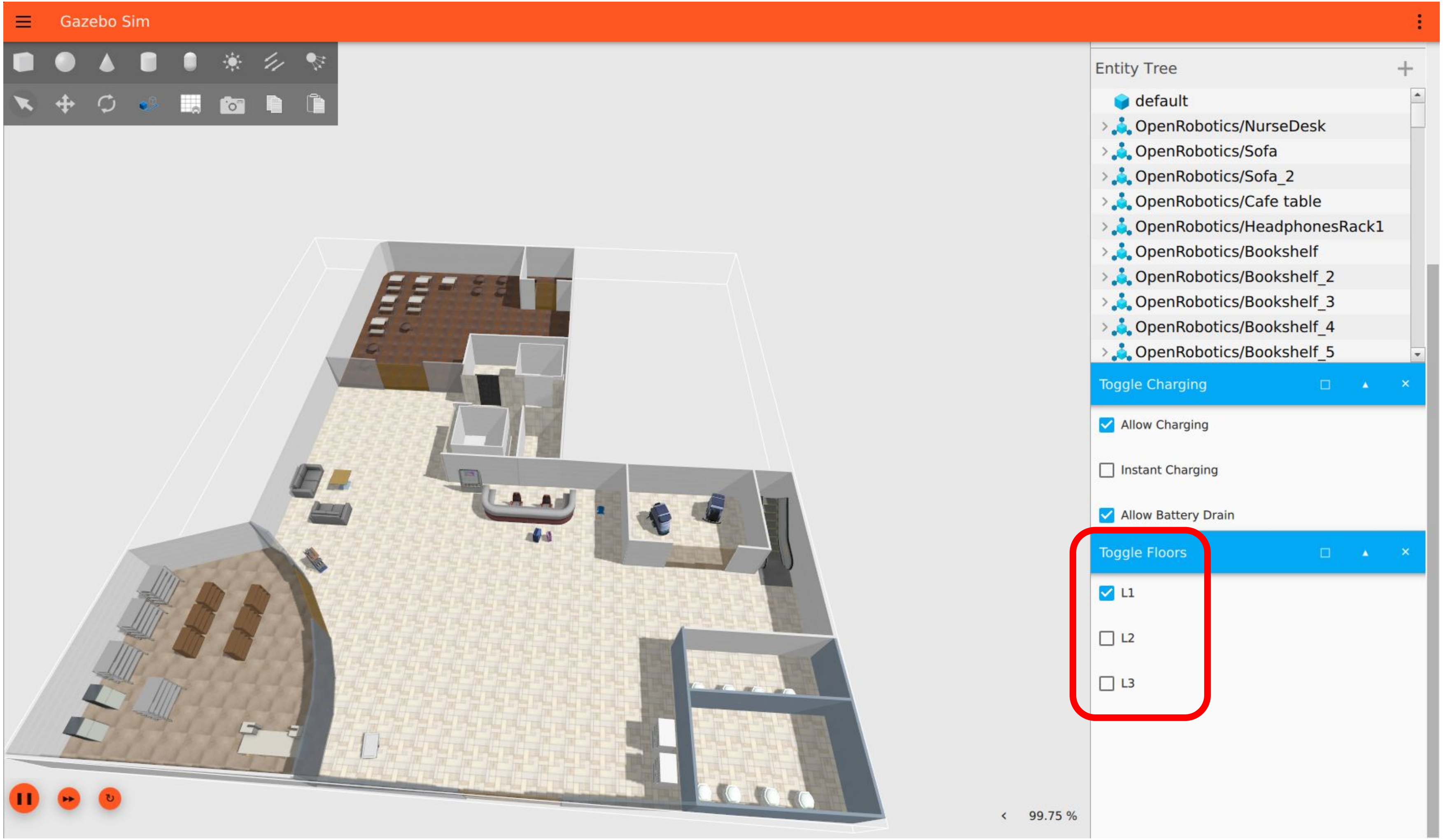

File Panels Help
Move Camera
Select
Focus Camera
Measure
2D Pose Estimate
2D Goal Pose
Publish Point
SchedulePanel
Negotiation id
Participants
RViz
Level selector
Output Topic:
Map Name: L3
Start Duration(s): 0
Query Du
600
(s)
SchedulePanel
LiftPanel
Reset
31 fps
#L3-01
#L3-02
#L3-03
#L3-04
#L3-05
#L3-06
#L3-07
#L3-08
#L3-09
#L3-10
Suite
#L3-11
#L3-12
#L3-13
#L3-14
#L3-15
E
maintenance
N
The Hotel
Level 3
*RMF Sample Floorplan
*Not to Scale


Gazebo Sim
World
Entity 1
+ System Plugin Info
+ Magnetic Field
Name sim_world
Scene
Physics Engine Plugin ...-dartsim-plugin
Atmosphere
Physics Collision Detector ode
Render Engine Gui Plugin
Render Engine Server Plugin
Physics Solver DantzigBoxedLcpSolver
+ Physics
World Sdf
Entity Tree
default
OpenRobotics/NurseDesk
OpenRobotics/Sofa
OpenRobotics/Sofa_2
OpenRobotics/Cafe table
OpenRobotics/HeadphonesRack1
OpenRobotics/Bookshelf
OpenRobotics/Bookshelf_2
OpenRobotics/Bookshelf_3
OpenRobotics/Bookshelf_4
OpenRobotics/Bookshelf_5
Toggle Charging
Allow Charging
Instant Charging
Allow Battery Drain
99.76 %

Gazebo


Gazebo Sim
Entity Tree
default
OpenRobotics/NurseDesk
OpenRobotics/Sofa
OpenRobotics/Sofa_2
OpenRobotics/Cafe table
OpenRobotics/HeadphonesRack1
OpenRobotics/Bookshelf
OpenRobotics/Bookshelf_2
OpenRobotics/Bookshelf_3
OpenRobotics/Bookshelf_4
OpenRobotics/Bookshelf_5
Toggle Charging
Allow Charging
Instant Charging
Allow Battery Drain
Toggle Floors
L1
L2
L3
99.74 %


Gazebo Sim
Entity Tree
default
OpenRobotics/NurseDesk
OpenRobotics/Sofa
OpenRobotics/Sofa_2
OpenRobotics/Cafe table
OpenRobotics/HeadphonesRack1
OpenRobotics/Bookshelf
OpenRobotics/Bookshelf_2
OpenRobotics/Bookshelf_3
OpenRobotics/Bookshelf_4
OpenRobotics/Bookshelf_5
Toggle Charging
Allow Charging
Instant Charging
Allow Battery Drain
Toggle Floors
L1
L2
L3
99.74 %


Gazebo Sim
Entity Tree
default
OpenRobotics/NurseDesk
OpenRobotics/Sofa
OpenRobotics/Sofa_2
OpenRobotics/Cafe table
OpenRobotics/HeadphonesRack1
OpenRobotics/Bookshelf
OpenRobotics/Bookshelf_2
OpenRobotics/Bookshelf_3
OpenRobotics/Bookshelf_4
OpenRobotics/Bookshelf_5
Toggle Charging
Allow Charging
Instant Charging
Allow Battery Drain
Toggle Floors
L1
L2
L3
99.75 %

Terminal 2

Here, we will showcase 2 types of Tasks: Patrol and Clean, you can dispatch them via CLI as follows:

```
docker run --rm -it \
  ghcr.io/open-rmf/rmf/rmf_demos:jazzy-rmf-latest \
    ros2 run rmf_demos_tasks dispatch_patrol \
      -p restaurant  L3_master_suite \
      -n 1 --use_sim_time
```

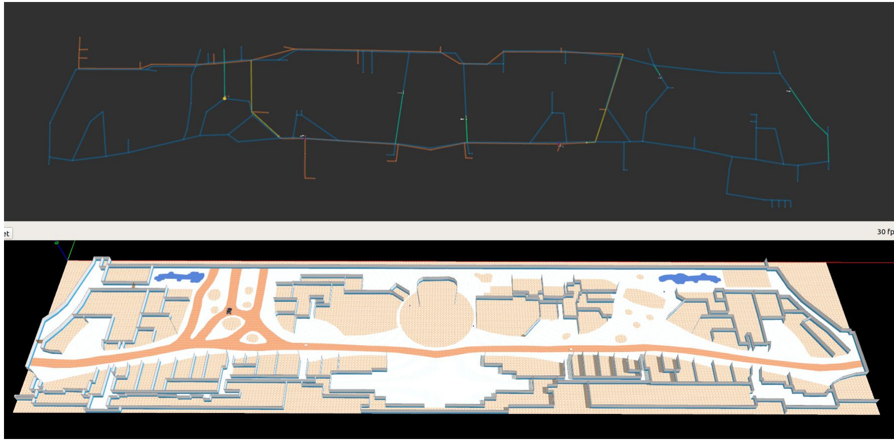
Terminal 2

Here, we will showcase 2 types of Tasks: Patrol and Clean, you can dispatch them via CLI as follows:

```
docker run --rm -it \
  ghcr.io/open-rmf/rmf/rmf_demos:jazzy-rmf-latest \
    ros2 run rmf_demos_tasks dispatch_clean \
    -cs clean_lobby --use_sim_time
```

Airport Terminal World

This demo world shows robot interaction on a much larger map, with a lot more lanes, destinations, robots and possible interactions between robots from different fleets, robots and infrastructure, as well as robots and users.



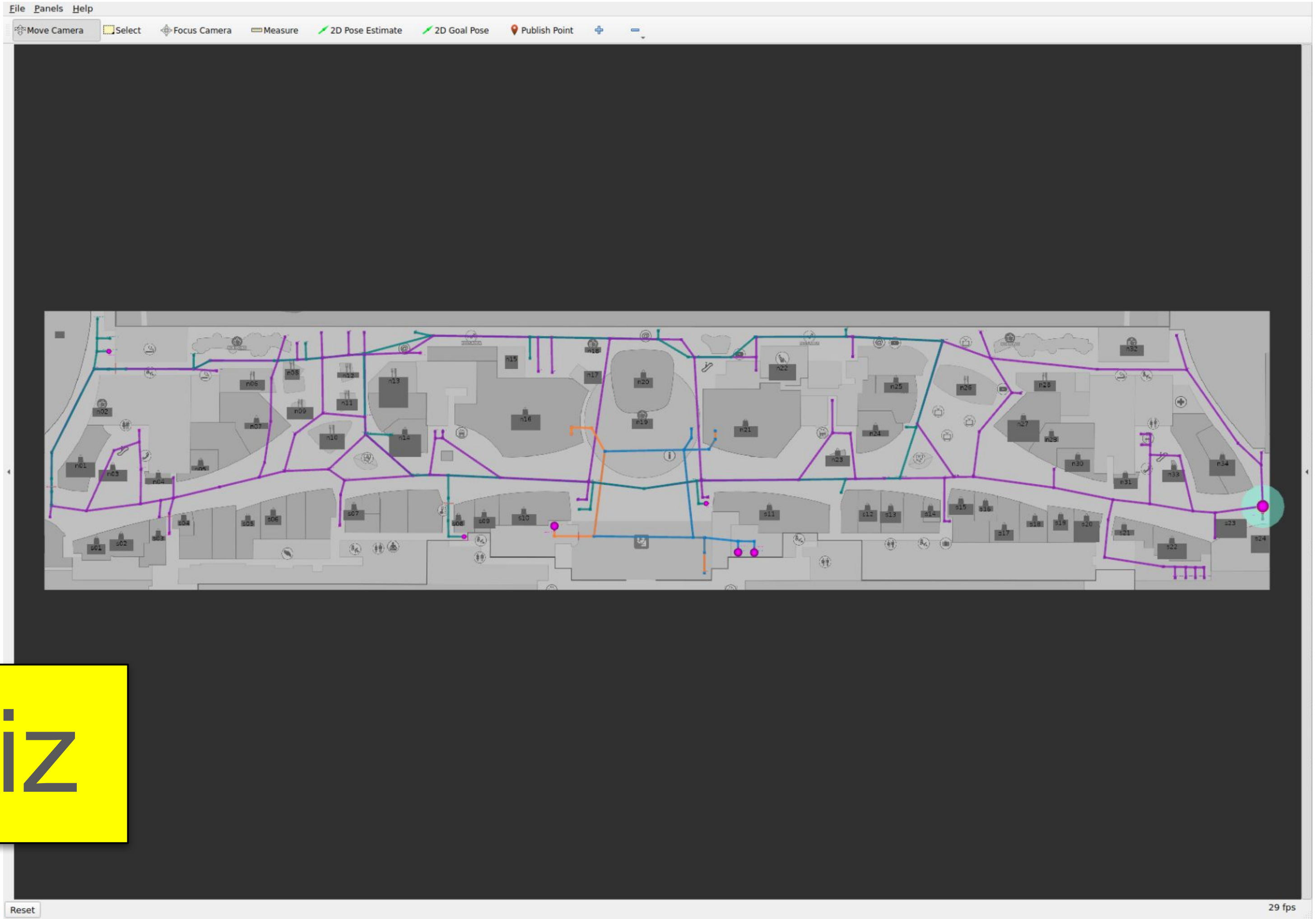
Terminal 1

```
rocker --nvidia --x11 \
  ghcr.io/open-rmf/rmf/rmf_demos:jazzy-rmf-latest \
    ros2 launch rmf_demos_gz \
      airport_terminal.launch.xml
```

Terminal 1

To see crowd simulation in action, enable crowd sim by:

```
rocker --nvidia --x11 \
  ghcr.io/open-rmf/rmf/rmf_demos:latest \
    ros2 launch rmf_demos_gz \
      airport_terminal.launch.xml \
      use_crowdsim:=1
```



File
Panels
Help
Move Camera
Select
Focus Camera
Measure
2D Pose Estimate
2D Goal Pose
Publish Point
Reset
29 fps

RViz


Gazebo Sim
Model
Entity 1987
Name
airport_terminal_L1
Wind Mode
Pose
Static
Model Sdf
Parent Entity
1
Source File Path
...erminal_L1/model.sdf
Model Canonical Link
Self Collide
Entity Tree
default
Fountain
Pine Tree
Pine Tree_2
Pine Tree_3
MainTable
Standing person
Standing person_2
Standing person_3
Standing person_4
Standing person_5
Toggle Charging
Allow Charging
Instant Charging
Allow Battery Drain

Gazebo

Terminal 2

You can submit patrol, delivery or clean task via CLI:

```
docker run --rm -it \
  ghcr.io/open-rmf/rmf/rmf_demos:jazzy-rmf-latest \
    ros2 run rmf_demos_tasks dispatch_patrol \
      -p s07 n12 \
      -n 3 --use_sim_time
```

Terminal 2

You can submit patrol, delivery or clean task via CLI:

```
docker run --rm -it \
  ghcr.io/open-rmf/rmf/rmf_demos:jazzy-rmf-latest \
    ros2 run rmf_demos_tasks dispatch_delivery \
      -p mopcart_pickup -ph mopcart_dispenser \
      -d spill -dh mopcart_collector \
      --use_sim_time
```

Terminal 2

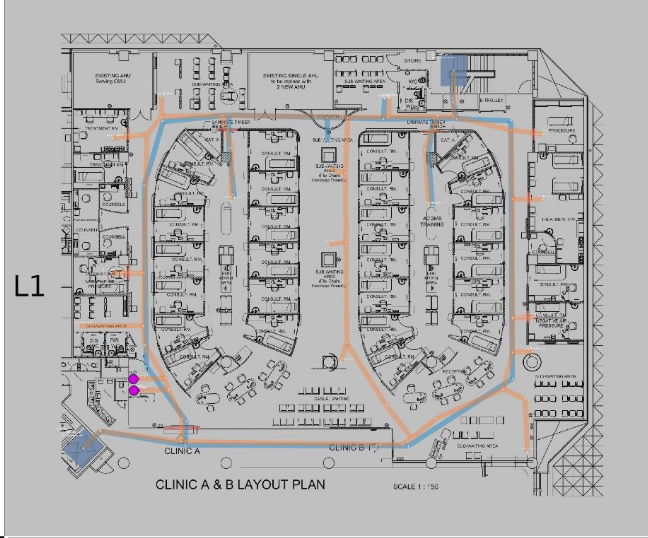
You can submit patrol, delivery or clean task via CLI:

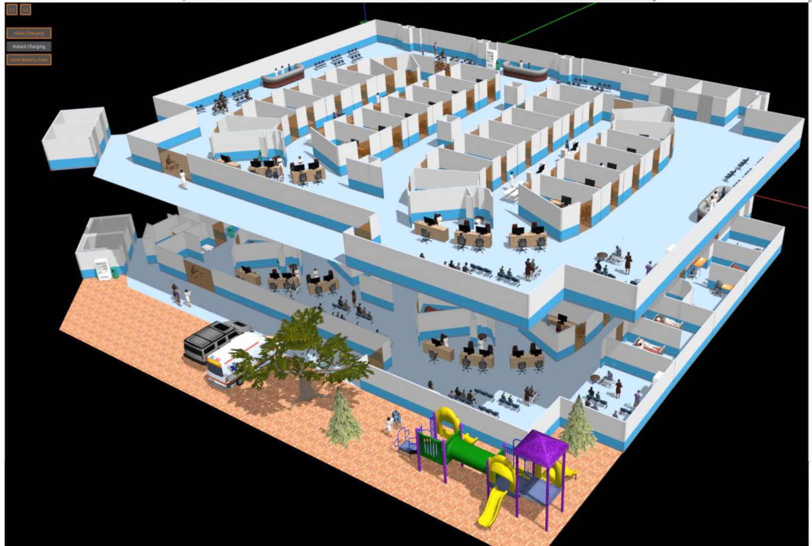
```
docker run --rm -it \
  ghcr.io/open-rmf/rmf/rmf_demos:jazzy-rmf-latest \
    ros2 run rmf_demos_tasks dispatch_clean \
      -cs zone_3 \
      --use_sim_time
```

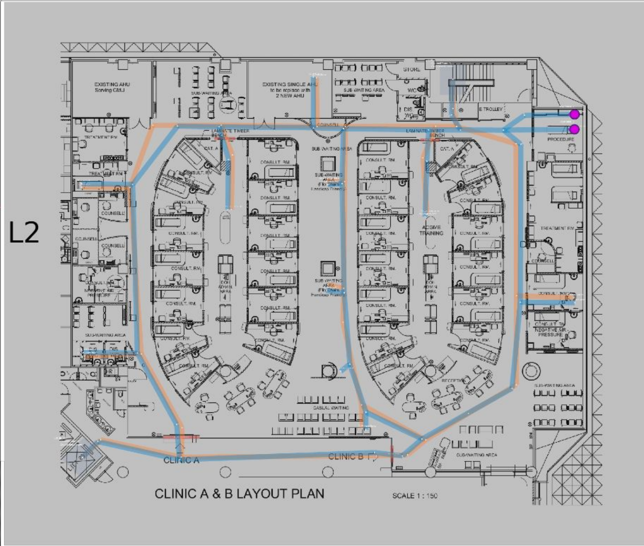
Clinic World

This is a clinic world with two levels and two lifts for the robots.

Two different robot fleets with different roles navigate across two levels by lifts.

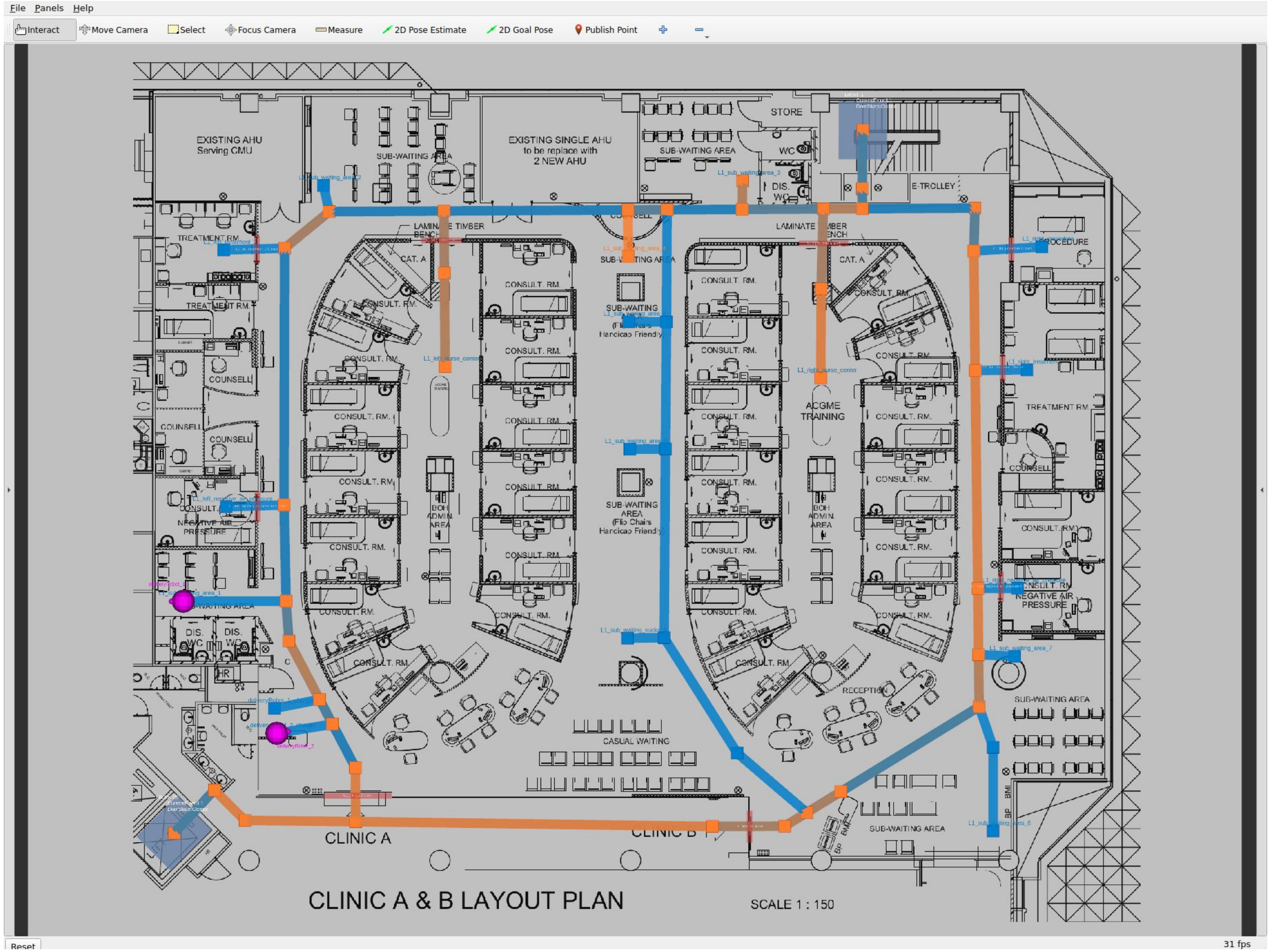






Terminal 1

```
rocker --nvidia --x11 \
  ghcr.io/open-rmf/rmf/rmf_demos:jazzy-rmf-latest \
    ros2 launch rmf_demos_gz \
      clinic.launch.xml
```


File Panels Help
Interact
Move Camera
Select
Focus Camera
Measure
2D Pose Estimate
2D Goal Pose
Publish Point
EXISTING AHU Serving CMU
SUB-WAITING AREA
EXISTING SINGLE AHU to be replace with 2 NEW AHU
STORE
WC
E-TROLLEY
TREATMENT RM.
CONSULT. RM.
COUNSELL
SUB-WAITING AREA (Flip Chairs Handicap Friendly)
ACGME TRAINING
BOH ADMIN AREA
NEGATIVE AIR PRESSURE
RECEPTION
CASUAL WAITING
CLINIC A
CLINIC B
CLINIC A & B LAYOUT PLAN
SCALE 1 : 150
Reset
31 fps

RViz


Gazebo Sim
World
Entity 1
Render Engine Server Headless
+ Gravity
+ Magnetic Field
Name sim_world
Atmosphere
Physics Collision Detector ode
Physics Engine Plugin ...-dartsim-plugin
Render Engine Gui Plugin
Render Engine Server Plugin
Physics Solver DantzigBoxedLcpSolver
+ Physics
World Sdf
Entity Tree
default
OpenRobotics/PotatoChipChair
OpenRobotics/PotatoChipChair_2
OpenRobotics/PotatoChipChair_3
OpenRobotics/PotatoChipChair_4
OpenRobotics/PotatoChipChair_5
OpenRobotics/PotatoChipChair_6
OpenRobotics/PotatoChipChair_7
OpenRobotics/PotatoChipChair_8
OpenRobotics/PotatoChipChair_9
OpenRobotics/PotatoChipChair_10
Toggle Charging
Allow Charging
Instant Charging
Allow Battery Drain
96.68 %

Gazebo

Terminal 2

You can submit tasks via CLI:

```
docker run --rm -it \
  ghcr.io/open-rmf/rmf/rmf_demos:jazzy-rmf-latest \
    ros2 run rmf_demos_tasks dispatch_patrol \
      -p L1_left_nurse_center \
         L2_right_nurse_center -n 5 \
      --use_sim_time
```

Terminal 2

You can submit tasks via CLI:

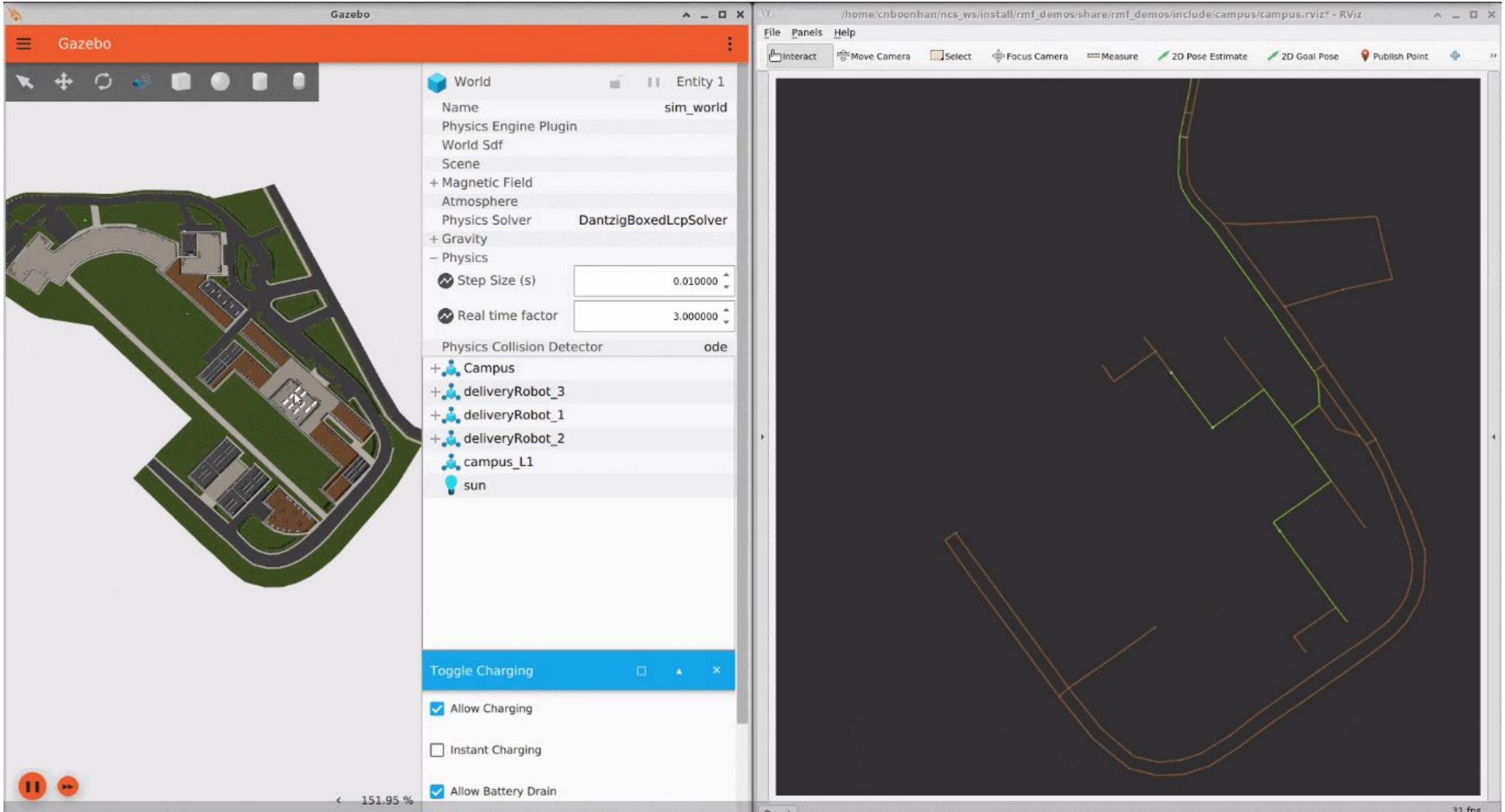
```
docker run --rm -it \
  ghcr.io/open-rmf/rmf/rmf_demos:jazzy-rmf-latest \
    ros2 run rmf_demos_tasks dispatch_patrol \
      -p L2_north_counter \
         L1_right_nurse_center -n 5 \
      --use_sim_time
```

Robots taking lift:

Multi-fleet demo:

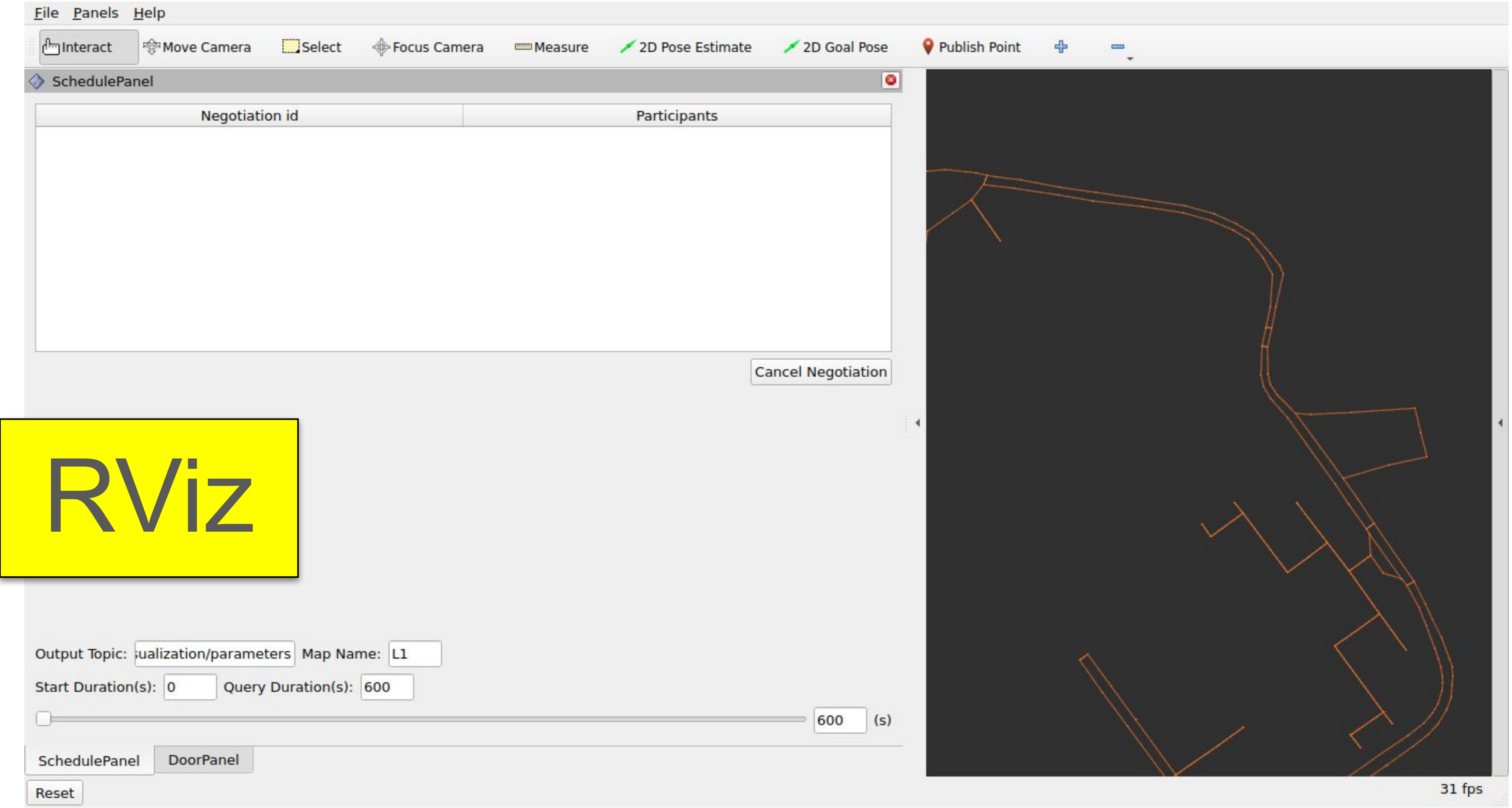
Campus World

This is a larger scale "Campus" World. In this world, there are multiple delivery robots that operate. The world is designed and traffic lanes are annotated at the planet scale, using GPS WGS84 coordinates.



Terminal 1

```
rocker --nvidia --x11 \
  ghcr.io/open-rmf/rmf/rmf_demos:jazzy-rmf-latest \
    ros2 launch rmf_demos_gz \
      campus.launch.xml
```


File Panels Help
Interact
Move Camera
Select
Focus Camera
Measure
2D Pose Estimate
2D Goal Pose
Publish Point
SchedulePanel
Negotiation id
Participants
Cancel Negotiation
RViz
Output Topic: ualization/parameters
Map Name: L1
Start Duration(s): 0
Query Duration(s): 600
600 (s)
SchedulePanel
DoorPanel
Reset
31 fps


Gazebo Sim
Model
Entity 4
Name
Campus
Static
Parent Entity
1
Model Sdf
Source File Path
...ls/Campus/model.sdf
+ Pose
Wind Mode
Model Canonical Link
Self Collide
Entity Tree
default
Campus
deliveryRobot_3
deliveryRobot_1
deliveryRobot_2
campus_L1
sun
Toggle Charging
Allow Charging
Instant Charging
Allow Battery Drain
99.44 %

Gazebo

Terminal 2

You can submit tasks via CLI:

```
docker run --rm -it \
  ghcr.io/open-rmf/rmf/rmf_demos:jazzy-rmf-latest \
    ros2 run rmf_demos_tasks dispatch_patrol \
      -p room_5 campus_4 -n 10 \
      --use_sim_time
```

Terminal 2

You can submit tasks via CLI:

```
docker run --rm -it \
  ghcr.io/open-rmf/rmf/rmf_demos:jazzy-rmf-latest \
    ros2 run rmf_demos_tasks dispatch_patrol \
      -p campus_5 room_3 \
      --use_sim_time
```

Terminal 2

You can submit tasks via CLI:

```
docker run --rm -it \
  ghcr.io/open-rmf/rmf/rmf_demos:jazzy-rmf-latest \
    ros2 run rmf_demos_tasks dispatch_patrol \
      -p room_2 dead_end -n 10 \
      --use_sim_time
```

Exercise

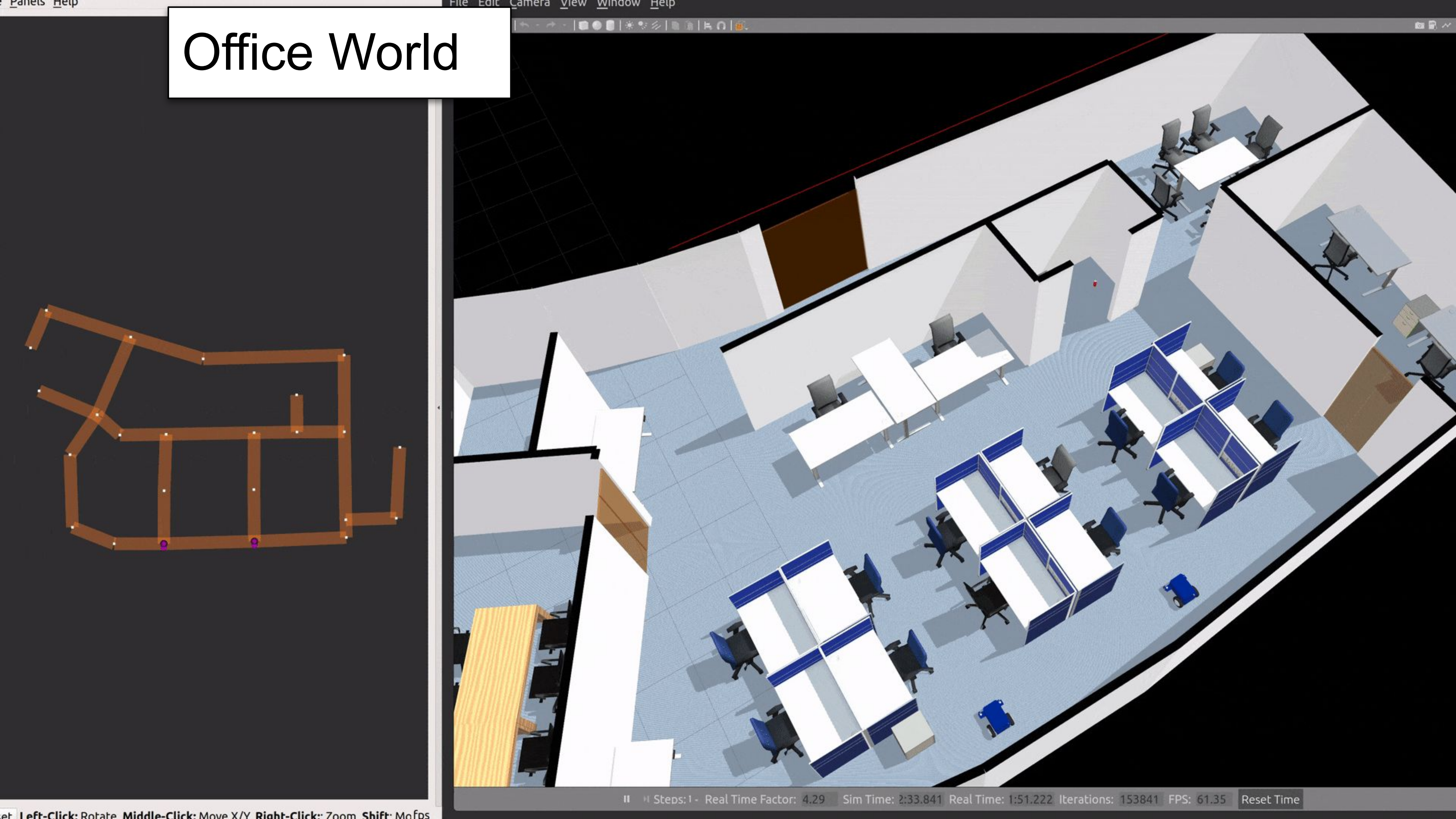
Simulate the office world

Dispatch two patrol tasks simultaneously

Record a bag of ROS topics from RViz

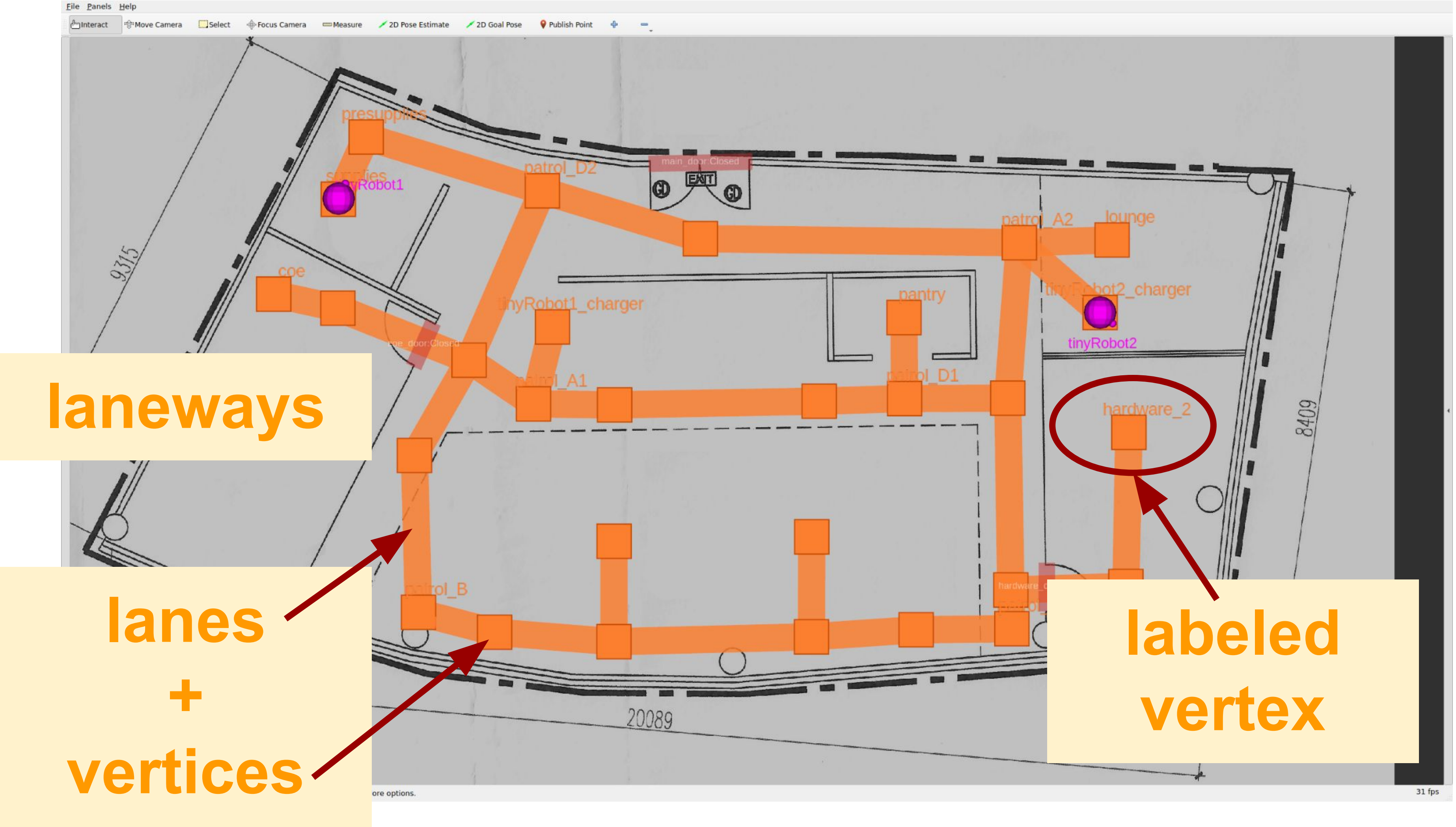
Record a screenshot of the Gazebo simulator

Office World



Terminal 1

```
rocker --nvidia --x11 \
  ghcr.io/open-rmf/rmf/rmf_demos:jazzy-rmf-latest \
    ros2 launch rmf_demos_gz \
      office.launch.xml
```


File Panels Help
Interact
Move Camera
Select
Focus Camera
Measure
2D Pose Estimate
2D Goal Pose
Publish Point
presupplies
patrol_D2
main_door:Closed
EXIT
patrol_A2
lounge
tinyRobot2_charger
tinyRobot2
pantry
tinyRobot1_charger
coe
coe_door:Closed
patrol_A1
patrol_D1
hardware_2
patrol_B
9315
8409
20089
31 fps
laneways
lanes
+
vertices
labeled
vertex

Terminal 2

You can submit tasks via CLI:

```
docker run --rm -it \
  ghcr.io/open-rmf/rmf/rmf_demos:jazzy-rmf-latest \
    ros2 run rmf_demos_tasks dispatch_patrol \
      -p <places_to_patrol_through> \
      -n <number_of_loops_to_perform> \
      --use_sim_time
```

rmf_demos/rmf_demos_ ×

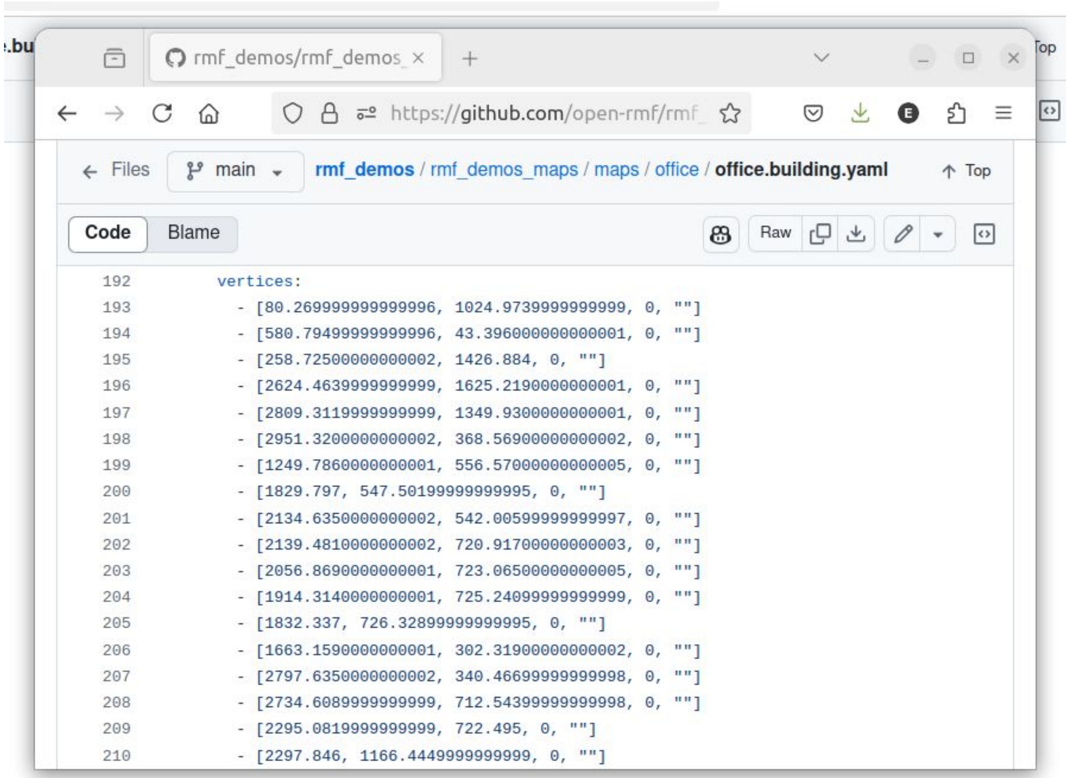
https://github.com/open-rmf/rmf_

← Files main rmf_demos / rmf_demos_tasks / rmf_demos_tasks / dispatch_patrol.py ↑ Top

Code Blame Raw

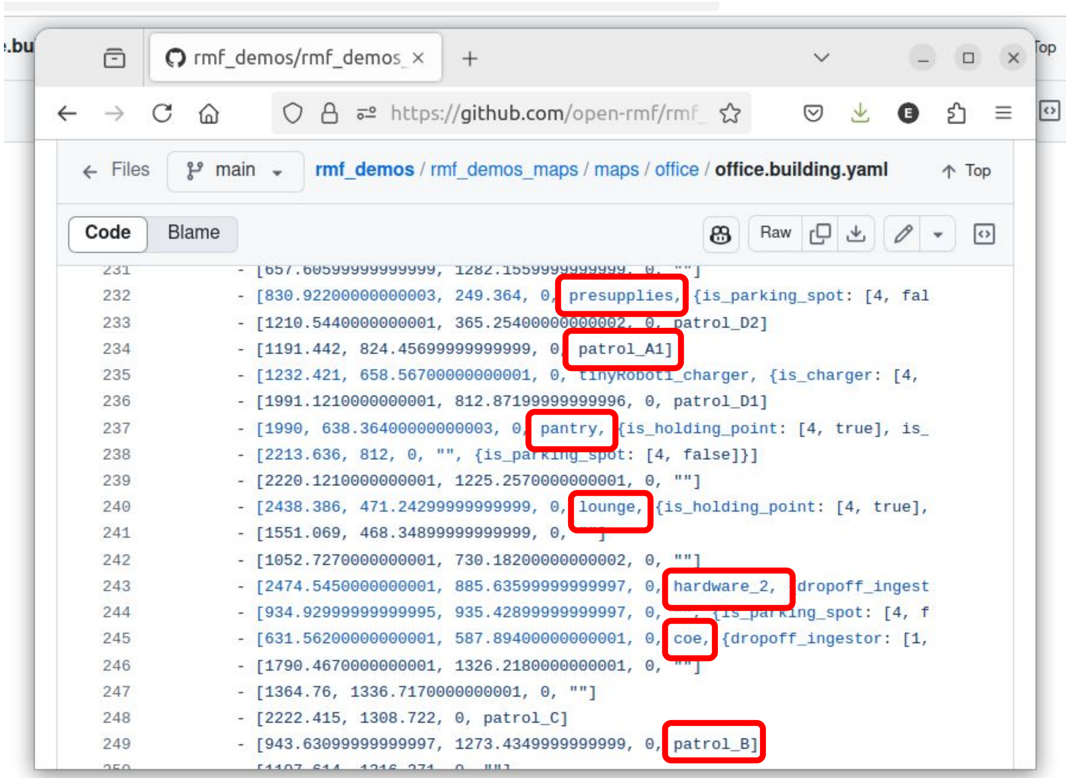
```python
class TaskRequester(Node):
    """Dispatch patrol task requester."""

    def __init__(self, argv=sys.argv):
        """Initialize task requester."""
        super().__init__('task_requester')
        parser = argparse.ArgumentParser()
        parser.add_argument(
            '-p',
            '--places',
            required=True,
            nargs='+',
            type=str,
            help='Places to patrol through',
        )
        parser.add_argument(
            '-n',
            '--rounds',
            help='Number of loops to perform',
            type=int,
            default=1,
        )
```

Places to patrol through



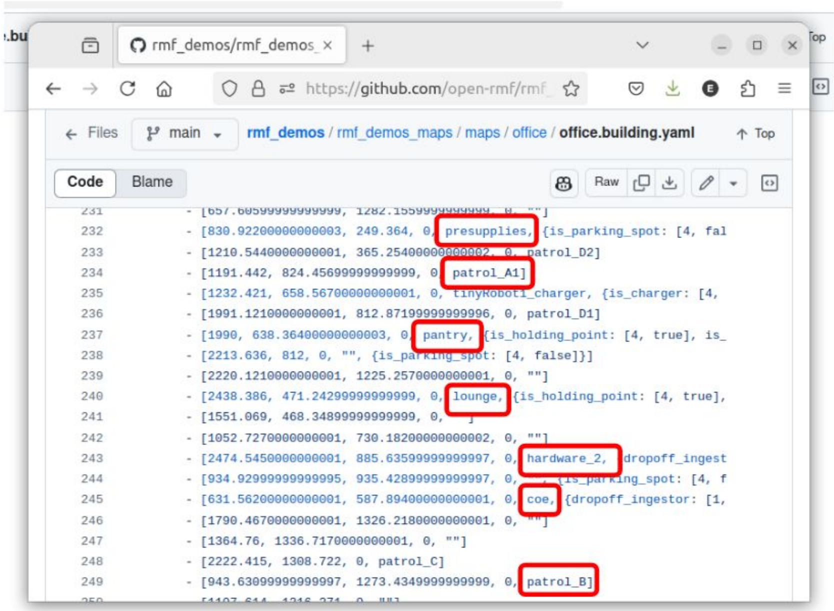

```yaml
        vertices:
          - [80.269999999999996, 1024.9739999999999, 0, ""]
          - [580.79499999999996, 43.396000000000001, 0, ""]
          - [258.72500000000002, 1426.884, 0, ""]
          - [2624.4639999999999, 1625.2190000000001, 0, ""]
          - [2809.3119999999999, 1349.9300000000001, 0, ""]
          - [2951.3200000000002, 368.56900000000002, 0, ""]
          - [1249.7860000000001, 556.57000000000005, 0, ""]
          - [1829.797, 547.50199999999995, 0, ""]
          - [2134.6350000000002, 542.00599999999997, 0, ""]
          - [2139.4810000000002, 720.91700000000003, 0, ""]
          - [2056.8690000000001, 723.06500000000005, 0, ""]
          - [1914.3140000000001, 725.24099999999999, 0, ""]
          - [1832.337, 726.32899999999995, 0, ""]
          - [1663.1590000000001, 302.31900000000002, 0, ""]
          - [2797.6350000000002, 340.46699999999998, 0, ""]
          - [2734.6089999999999, 712.54399999999998, 0, ""]
          - [2295.0819999999999, 722.495, 0, ""]
          - [2297.846, 1166.4449999999999, 0, ""]
```

Places to patrol through

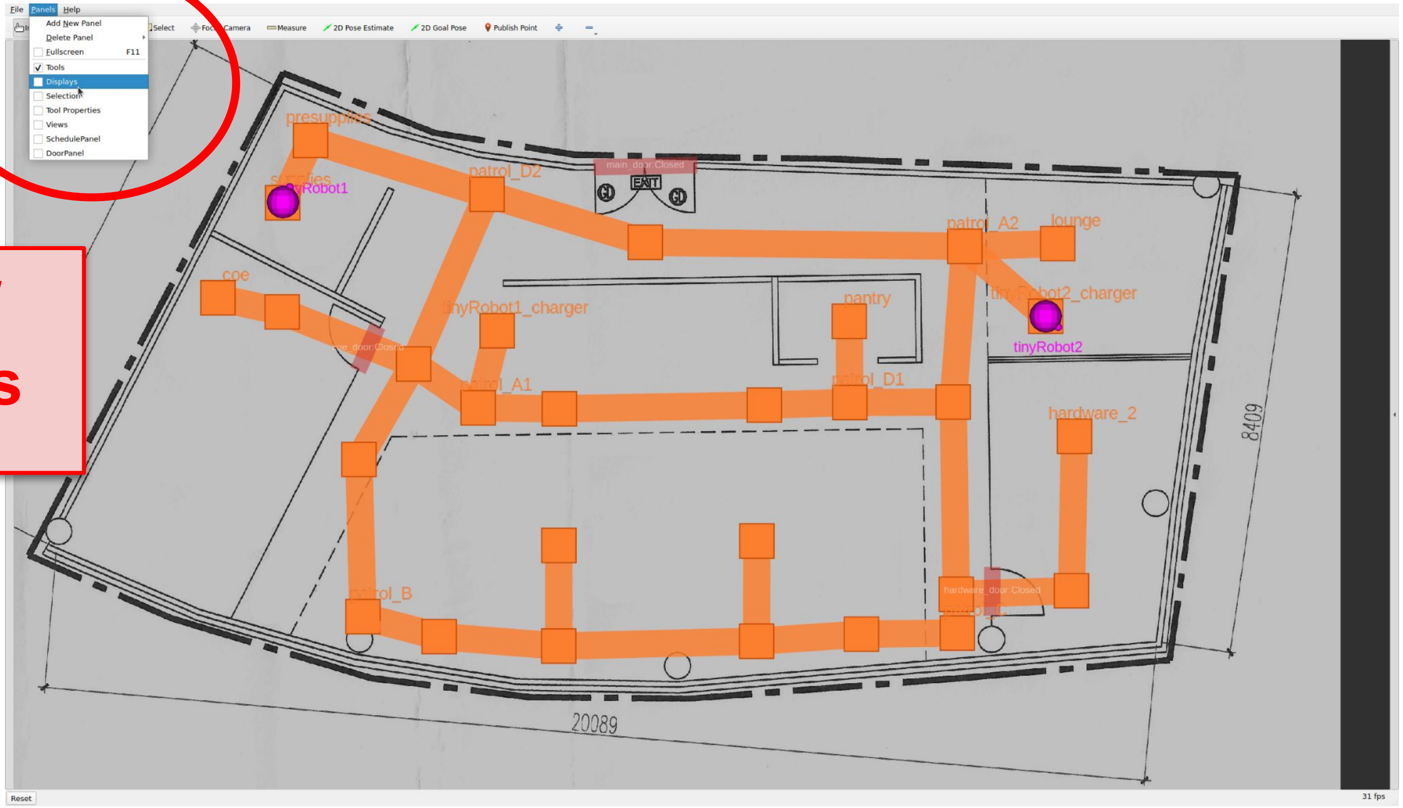


Places to patrol through

Select 3 places from the set of labeled vertices

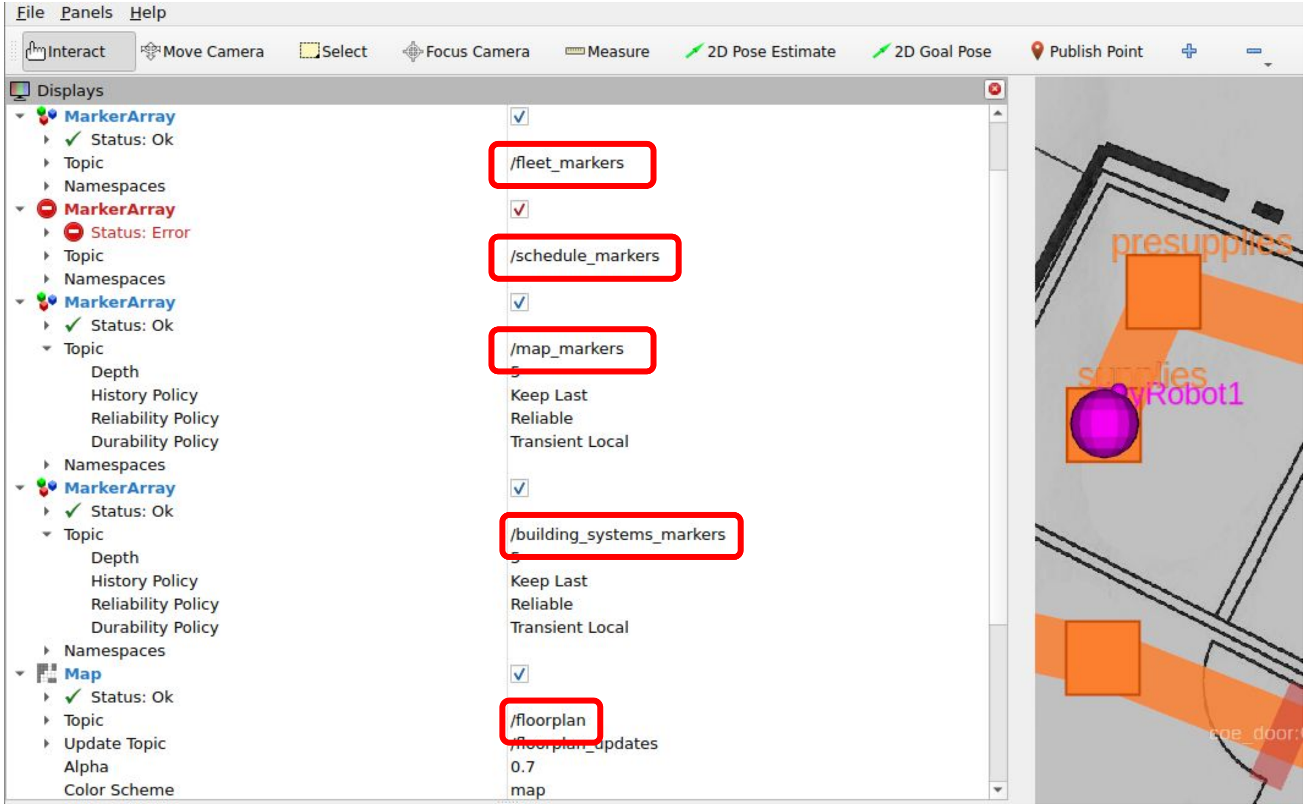


ROS topics from RViz



Panels / Displays

ROS topics from RViz



Terminal 3

Share a folder for recording the rosbags:

```
docker run --rm -it \
  -v `pwd`/rosbags:/rosbags \
  ghcr.io/open-rmf/rmf/rmf_demos:jazzy-rmf-latest \
    ros2 bag record -o /rosbags/office \
      /clock /fleet_markers /schedule_markers \
      /map_markers /building_systems_markers \
      /floorplan
```

Shared folder

Output file

Playing and visualizing the rosbag

First, stop the simulation!!!

Playing and visualizing the rosbag

```
docker run --rm -it \
  -v `pwd`/rosbags:/rosbags \
  ghcr.io/open-rmf/rmf/rmf_demos:jazzy-rmf-latest \
    ros2 bag play /rosbags/office
```

Shared folder

rosbag file

Playing and visualizing the rosbag

```
rocker --nvidia --x11 \
  ghcr.io/open-rmf/rmf/rmf_demos:jazzy-rmf-latest \
   "rviz2 -d
/rmf_demos_ws/install/rmf_demos/share/rmf_dem
os/include/office/office.rviz"
```

References

- Open-RMF demos
- Gazebo Harmonic
- ROS2
  - RViz User Guide
  - Marker: Display types
- OSRF/rocker
- Docker
  - Networking overview
  - Sharing local files with containers